

Muminul Islam

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Passport: A11143242 **Place of birth:** Tangail, Bangladesh **Nationality:** Bangladeshi

ABOUT ME

AI Researcher passionate about optimising machine learning models, building resilient systems, and engineering efficient solutions for real-world, resource-constrained challenges.

EDUCATION AND TRAINING

[01/05/2020 – 01/02/2025]

Bachelor of Science

United International University

City: Dhaka | **Country:** Bangladesh | **Field(s) of study:** Computer Science and Engineering | **Final grade:** 3.0/4.0 (~80%) | **Thesis:** Low-Resource Machine Translation (Regional Dialect to Standard Dialect) using Transformer models.

RESEARCH EXPERIENCE

[01/11/2025 – Current]

Academic Research Collaborator | Supervised by Dr. Md Adnan Arefeen

Co-Authored Publication: Contributed to "RE-SciVQA: Improving Scientific VQA with Context-Relevant Retrieval and Lightweight MLLM Adaptation," a paper currently under review at a Q1 scientific journal (Impact Factor: 6.8, CiteScore: 14.4).

Advanced Model Optimization: Utilized parameter-efficient Low-Rank Adaptation (QLoRA) to fine-tune 7B-parameter models (Llama-3, Qwen) on limited compute resources, achieving a Rouge score (R1-F1) of 73.9% and outperforming highly-resourced models.

Solved Scientific Retrieval Challenges: Developed a relevance-enhanced RAG pipeline with query expansion and plot digitization, improving exact match by 11.9% and fuzzy match by 29.5% compared to standard baselines.

Rigorous Evaluation: Established standardized benchmarking protocols for handling closed-ended infinite visual questions and complex scientific literature figures.

Link: https://dx.doi.org/10.2139/ssrn.6302829

HONOURS AND AWARDS

Champion at UIU Intra-University AI Contest

Task: Developed a multiclass classification model to predict medical conditions from research paper abstracts.

Links: https://www.facebook.com/share/p/17eSGLCg4K/ | https://www.kaggle.com/competitions/uiu-intra-university-ai-contest-fall-2022/leaderboard

Merit Scholarship Awarding institution: United International University

Awarded twice for consistent academic excellence at United International University.

Link: https://drive.google.com/file/d/14t_KClJCCohVJICuYDNtn6BveB4bhcFL/view?usp=sharing

3rd Runner-up at System Analysis and Design Lab Project Show

Selected among 100+ participants

Link: https://drive.google.com/file/d/1Vp_gVhaYJPx7sTP753SbbgsGESmXqNiq/view?usp=sharing

Contestant at International Collegiate Programming Contests(ICPC)

Asia Dhaka Regional: Rank **135/309** (Onsite, 2024);

Honourable Mention: 2021, 2022, 2023

Link: https://drive.google.com/drive/folders/19BgR-S0QnDhtqxHJJn0GPoPZTD8w7Q4a?usp=sharing

Contestant at Inter University Programming Contests

Rank 7 - UAP Collaborative Contest 2024;

Rank 50 - KUET IUPC 2025;

Rank 52 - SUST IUPC 2024;

Rank 53 - IUT IUPC 2024.

(Out of 130+ best teams from each university)

Link: https://drive.google.com/drive/folders/1Fu0IqnCgcWxA9CNo6KKOz1Dkvij1niex

PROJECTS

E-Commerce Inventory API

Delivered a secure .NET 8 inventory API with JWT/BCrypt authentication, polished product/category CRUD (pagination, search, safe deletes), repository-based data access, and deployment-ready PostgreSQL Server support.

Link: https://github.com/mijami/ecommerce-inventory-api

Codewar

Competitive programming platform with practice problems and analytics dashboards

Link: https://github.com/mijami/codewar

LinguaLearner

Cross-platform language learning app with quizzes, blogs, and secure authentication.

Tech: React Native (Expo), Express.js, TypeScript, MongoDB

Link: https://github.com/mijami/LinguaProject

SKILLS

Problem Solving
C | C++ | DSA(Data Structure and Algorithms)
AI/ML
Python (NumPy, Pandas, NLTK, Matplotlib, sklearn, TensorFlow) | SQL
Software Engineering
Git | TypeScript | Object Oriented Programming | MVC Architecture

LANGUAGE SKILLS

Mother tongue(s): Bengali

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

VOLUNTEERING

Member UIU Competitive Programming Community

Link: https://www.facebook.com/profile.php?id=61568564875384

HOBBIES AND INTERESTS

Competitive Programming

Passionate about solving complex algorithmic challenges, consistently practising data structures and algorithms. Regularly participate in international online contests (Codeforces) to sharpen analytical thinking and problem-solving efficiency.

Link: https://codeforces.com/profile/b4tman

RECOMMENDATIONS

Name: Md Adnan Arefeen Research Supervisor

Assistant Professor at North South University.

Email: adnan.arefeen@northsouth.edu

Link: https://scholar.google.com/citations?user=szq4caMAAAAJ&hl=en

Name: Md. Tarek Hasan ICPC Team Coach

Assistant Professor at United International University

Email: tarek@cse.uiu.ac.bd

Link: https://scholar.google.com/citations?user=SFfUSOIAAAAJ&hl=en